MALAWI NATIONAL EXAMINATIONS BOARD

2023 MALAWI SCHOOL CERTIFICATE OF EDUCATION (MSCE) EXAMINATION

BIOLOGY - PAPER II (THEORY)

Subject Code: M021/II | Time Allowed: 2 Hours

INSTRUCTIONS TO CANDIDATES:

1. Section A: Answer ALL questions in the spaces provided (60 Marks).
2. Section B: Answer any TWO questions in essay form (40 Marks).
3. Diagrams should be drawn in pencil with clear labels.

CANDIDATE IDENTIFICATION:

Candidate Examination Number: [_____]
Examination Centre Name: [_____]

SECTION A: CELL BIOLOGY & PHYSIOLOGY

Question 1: Photosynthesis and Cellular Respiration

(a) Differentiate between the light and dark stages of photosynthesis. [4 marks]

(b) State the role of magnesium ions in chlorophyll synthesis. [2 marks]

(c) Describe the mechanism of gaseous exchange in dicotyledonous leaves. [4 marks]

Question 2: Genetics and Inheritance

(a) Define Mendel's Law of Independent Assortment. [2 marks]

(b) In a monohybrid cross between two heterozygous tall pea plants (Tt), determine the phenotypic and genotypic ratios of the F1 generation. [6 marks]